

Motov - MotorBike rental management system

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Abstract. The rapid growth of digital platforms has transformed the transportation and rental industry. However, many motorbike rental businesses still rely on manual processes, resulting in inefficient booking management, vehicle tracking, identity verification, and payment handling. This paper proposes Motov, a hybrid motorbike rental management platform that integrates company-owned and owner-owned vehicles into a unified ecosystem. The system provides web and mobile applications supporting customers, staff, vehicle owners, and administrators. Core functionalities include vehicle catalog management, online booking, rental conflict validation, identity verification, payment processing, notification services, and administrative analytics. The proposed solution aims to improve operational efficiency, increase vehicle utilization, enhance customer experience, and support secure rental transactions. The platform is implemented using ReactJS, React Native, ASP.NET Core Web API, and MongoDB.

Keywords: Motorbike Rental · Vehicle Management · Online Booking · Identity Verification · Mobile Application

1 Introduction

The demand for motorbike rental services has increased significantly due to tourism growth, urban mobility needs, and the popularity of short-term transportation solutions. In Vietnam, many rental businesses still operate through manual processes such as phone calls, social media messaging, and paper-based agreements.

Traditional rental approaches often face several challenges, including inefficient vehicle management, limited booking transparency, and difficulties in tracking rental transactions. Furthermore, customers may experience inconvenience when searching for available vehicles, while rental businesses struggle to maximize vehicle utilization.

Recent studies have investigated vehicle rental systems, fleet management, customer satisfaction, and peer-to-peer vehicle sharing models [3, 2, 5, 4, 6, 7].

This project proposes Motov, a hybrid motorbike rental platform that supports both company-owned and partner-owned vehicles. The system allows customers to search and book vehicles online, partners to register and manage their vehicles, and administrators to oversee platform operations.

The main contributions of this project are:

- Development of a hybrid rental model integrating company-owned and owner-owned motorbikes.
- Design of a centralized rental management platform supporting customers, staff, owners, and administrators.
- Integration of identity verification, booking conflict validation, and automated rental price calculation mechanisms.
- Implementation of web and mobile applications using modern full-stack technologies.
- Improvement of booking transparency, operational efficiency, and vehicle utilization.

2 Related Work

Several studies have investigated vehicle rental, fleet management, customer satisfaction, and car-sharing systems.

Khawaja and Yang [3] examined the relationship between employee engagement, customer satisfaction, and customer retention in the car rental industry. Their findings indicated that service quality significantly influences customer loyalty and business performance. However, the study focused on customer behavior rather than software platform development.

Gonzalez et al. [2] proposed a fleet utilization metric for shared mobility systems. Although the study improved operational evaluation, it did not address booking management or payment integration.

Munzel et al. [5] analyzed different car-sharing business models and demonstrated the viability of peer-to-peer vehicle sharing. However, the research did not provide a practical software implementation.

Liu et al. [4] investigated booking optimization algorithms for online car-sharing systems. Their work improved vehicle allocation efficiency but lacked user management and payment functionalities.

Neifer et al. [6] explored trust-building mechanisms in peer-to-peer car-sharing platforms. The study highlighted the importance of reputation systems but did not provide a complete rental management solution.

Andersen [1] proposed a queueing model for rental scheduling and inventory optimization. However, the study focused on mathematical optimization rather than platform implementation.

Soppert et al. [7] studied the integration of traditional rental and car-sharing services. Their findings showed improved utilization and profitability, but no web or mobile implementation was proposed.

Although previous studies have focused on fleet utilization, scheduling optimization, customer satisfaction, and peer-to-peer sharing, few studies provide a comprehensive platform that integrates booking, payment processing, vehicle management, partner vehicle registration, and administrative monitoring.

Therefore, this project proposes a hybrid motorbike rental platform that combines company-owned and partner-owned vehicles through a centralized web and mobile ecosystem.

3 Method

This study adopts the Design Science Research (DSR) approach to develop and evaluate the Motov platform. The objective is to design an artifact that addresses practical challenges in motorbike rental operations, including fragmented booking processes, inefficient vehicle utilization, and insufficient customer verification mechanisms.

3.1 Research Framework

The proposed research process consists of five stages:

1. Problem Identification: Investigating limitations of traditional motorbike rental businesses through requirement analysis and stakeholder discussions.
2. Objective Definition: Defining system goals related to booking transparency, operational efficiency, customer convenience, and security enhancement.
3. Artifact Design: Designing the hybrid rental platform architecture, database structure, and functional modules.
4. Development and Implementation: Building the web application, mobile application, backend services, and database components.
5. Evaluation: Verifying that the implemented functionalities satisfy predefined requirements and business rules.

3.2 Development Process

The Agile Scrum methodology was adopted to support iterative development and continuous stakeholder feedback. The development activities were divided into multiple sprints, each consisting of planning, implementation, testing, review, and retrospective phases.

This approach enabled the development team to continuously refine system functionalities and rapidly respond to requirement changes throughout the project lifecycle.

3.3 Proposed System Architecture

The proposed system adopts a three-tier architecture consisting of presentation, application, and data layers, as illustrated in Figure ??.

- Presentation Layer: ReactJS web application for administrators, staff, and owners, together with React Native mobile applications for customers.
- Application Layer: ASP.NET Core Web API responsible for business logic processing, authentication, booking validation, payment handling, and notification management.
- Data Layer: MongoDB database storing user profiles, vehicle information, bookings, payments, verification records, and system logs.

The layered architecture improves maintainability, scalability, and separation of concerns while supporting future system expansion.

3.4 Hybrid Rental Model

Unlike conventional rental systems that only support company-owned fleets, Motov introduces a hybrid rental model integrating both company-owned and owner-owned vehicles.

The proposed model allows individual vehicle owners to register their motor-bikes on the platform while administrators maintain centralized supervision over operational activities. This approach increases fleet availability without requiring proportional investments in vehicle acquisition.

The hybrid model provides benefits for multiple stakeholders:

- Customers gain access to a wider range of vehicles.
- Vehicle owners receive an additional income source.
- Administrators maintain unified management and quality control.
- The platform improves overall vehicle utilization rates.

3.5 Booking and Validation Mechanism

The booking workflow constitutes the core process of the proposed system.

When a customer submits a booking request, the system executes the following sequence:

1. Verify customer authentication status.
2. Validate customer identity verification status.
3. Check vehicle availability during the requested rental period.
4. Detect scheduling conflicts with existing reservations.
5. Calculate the rental price according to duration and promotional policies.
6. Process online or cash payment options.
7. Generate booking records and update vehicle status.
8. Dispatch confirmation notifications to relevant stakeholders.

This mechanism minimizes double-booking incidents and ensures consistency across rental transactions.

3.6 Rental Conflict Validation

To prevent overlapping reservations, the system employs a temporal validation mechanism.

For a requested booking interval (S_r, E_r) and an existing booking interval (S_e, E_e) , a conflict occurs if:

$$S_r < E_e \wedge E_r > S_e$$

If the condition evaluates to true, the booking request is rejected and alternative scheduling options must be selected.

This validation strategy guarantees that each vehicle can only participate in one active rental transaction during a specific time interval.

3.7 Rental Price Calculation

The system automatically computes rental costs using the following formula:

$$TotalPrice = (RentalDays \times DailyRate) - Discount + AdditionalFees$$

where:

- *RentalDays* represents the number of rental days;
- *DailyRate* denotes the vehicle rental fee per day;
- *Discount* refers to promotional reductions;
- *AdditionalFees* include late-return penalties or supplementary service charges.

Automated price calculation reduces human errors and improves pricing transparency.

3.8 Identity Verification Process

To improve transaction security, Motov incorporates a citizen identity verification process.

Customers are required to upload images of their Citizen Identification Card (CCCD). Administrators review submitted documents before granting verified status. Only verified customers are eligible to complete vehicle pickup procedures.

This mechanism enhances trust among stakeholders and reduces risks associated with fraudulent transactions.

3.9 Evaluation Strategy

The proposed platform is evaluated using functional testing methods based on predefined use cases and business requirements.

The evaluation focuses on the following criteria:

- Correctness of booking operations;
- Accuracy of rental conflict detection;
- Reliability of price calculation mechanisms;
- Effectiveness of identity verification workflows;
- Usability of web and mobile interfaces.

Successful completion of these evaluation criteria indicates that the proposed solution effectively addresses the operational limitations of conventional motorbike rental systems.

4 System Analysis

4.1 Problem Analysis

Many motorbike rental businesses still depend on manual workflows, including phone-based reservations, social media communication, and paper-based contracts. These approaches often lead to inefficient booking management, poor vehicle utilization, booking schedule conflicts, lack of customer identity verification, and limited operational visibility. In addition, rental businesses face difficulties in monitoring fleet availability, tracking rental transactions, and generating business reports.

4.2 Proposed Solution

The proposed solution is a centralized hybrid motorbike rental platform that supports both company-owned and owner-owned vehicles. The system provides web and mobile applications for customers, owners, staff, and administrators. It integrates online booking, payment processing, citizen identity verification, notification services, rental conflict validation, inventory management, and business analytics into a single ecosystem.

5 System Design

5.1 Use Case Design

Main actors:

- **Admin:** Manages the entire system including users, vehicles, reports, and platform settings. Admin is a specialized role of Customer.
- **Guest:** Unregistered user who can browse the website, search motorbikes, and view vehicle information without logging in.
- **Customer:** Registered user who can book motorbikes, make payments, manage bookings, and submit reviews.
- **Owner:** Manages rental shop information, motorbike inventory, pricing, and rental operations. Owner is a specialized role of Customer.
- **Staff:** Supports daily rental activities such as confirming bookings, checking vehicle conditions, and handling returns. Staff is a specialized role of Customer.

The use case diagrams for each actor are presented in Figures 1–6.

5.2 Database Design

The system architecture relies on a robust database schema encompassing several key entities to manage users, vehicles, bookings, and transactions efficiently:

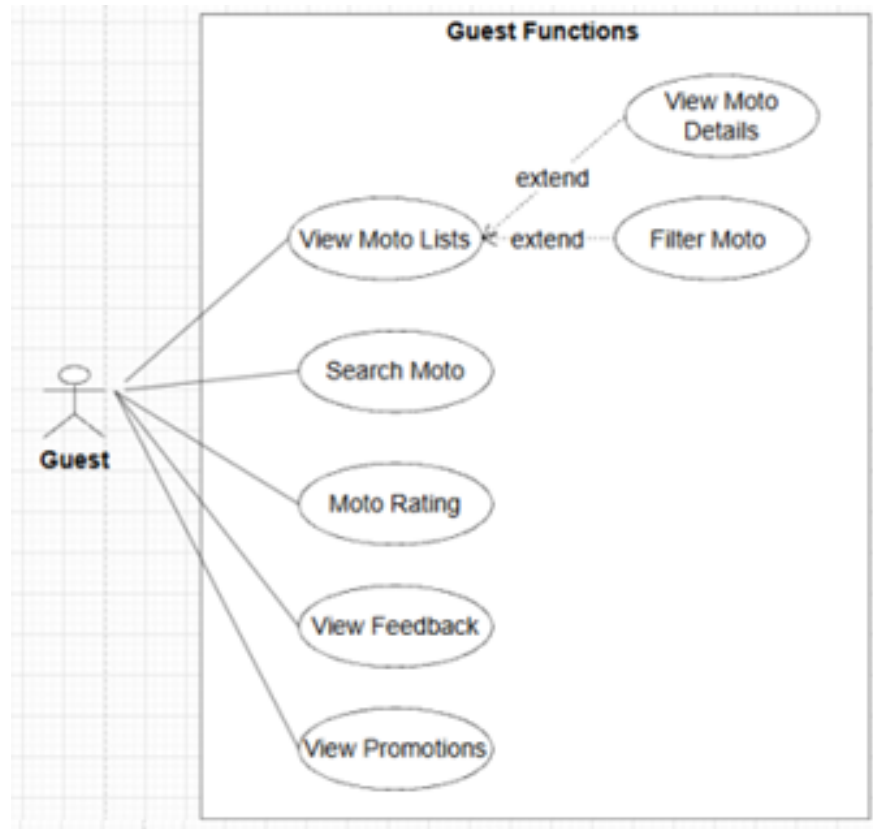


Fig. 1. Guest Use Case Diagram

- **Users:** Stores identity, authentication, profile, and security status for all system users (including customers and vehicle owners).
- **UserDocuments:** Manages legal verification documents uploaded by users, such as National IDs, passports, or driver's licenses.
- **Vehicle:** Contains detailed specifications, ownership, pricing, and current availability status of vehicles listed on the platform.
- **VehicleCategories:** Classifies vehicles into distinct structural types (e.g., Sedan, SUV, Hatchback, Truck) for filtering and pricing strategies.
- **TransmissionTypes:** Defines the operational transmission systems of the vehicles (e.g., Manual, Automatic).
- **VehicleImages:** Stores media assets and image paths linked to listed vehicles for marketplace display.
- **VehicleFeatures:** Maps specific features and onboard amenities (e.g., GPS, Sunroof, Dashcam, Bluetooth) to vehicles.
- **Booking:** Captures the core transactional lifecycle of a vehicle rental reservation, including rental durations, base costs, and order statuses.

- **BookingApproval:** Records the audit log and approval state workflow executed by owners or system administrators for pending bookings.
- **Contract:** Manages digital legal agreements and e-contracts generated between renters and the platform/owners upon booking confirmation.
- **Payment:** Logs financial transaction details, billing metrics, and payment gate statuses for deposits and rental charges.
- **VehicleConditionLogs:** Tracks the physical state of the vehicle (fuel levels, odometer readings, scratches, damages) during pick-up and drop-off checklists.

Figure 7 presents the Entity Relationship Diagram (ERD) of the proposed system.

5.3 Screen Descriptions and Authorization

The application provides specialized screens tailored for different actor roles, ensuring secure and restricted access to platform functionalities:

- **Login & Registration Screen:** Provides form fields for email/password authentication, password recovery triggers, and single-click integration hooks for Google OAuth SSO. Available to all users.
- **Motorbike Catalog Portal:** A primary grid layout displaying available fleet vehicles, integrating text search boxes, multi-attribute filter widgets (brand, category), and sorting dropdowns. Accessible by Guests, Customers, Staff, and Admins.
- **Motorbike Details View:** Displays granular specifications of a chosen vehicle, customer aggregate scores/reviews, dynamic related bike carousel recommendations, and action toggles to add to wishlists or initiate booking. Accessible by all roles.
- **Booking Checkout Desk:** A multi-step form workflow where customers select dates, verify vehicle availability, upload a Citizen ID file attachment, and select the payment type. Restricted to Customers.
- **Customer Booking List:** A personalized dashboard for customers to review their rental histories, track active booking lifecycle stages, or trigger cancellations. Restricted to Customers.
- **Daily Rental Board:** A spreadsheet-style operational desk enabling Staff to monitor daily pickup/return schedules, input cash handovers, and change bike statuses. Restricted to Staff and Admins.
- **Admin Analytics Dashboard:** An executive control panel rendering real-time KPI data charts, automated low availability alerts, and data tables for user accounts, promotions, and fleet inventory management. Restricted to Admins.

6 Implementation

6.1 System Code Packages and Modules

The system’s functional architecture is modularized into dedicated code packages to maintain separation of concerns:

- **member_authority & slide_authority**: Handles user identity classification, session validation (JWT), and authorization control across restricted routes.
- **registration**: Coordinates the onboarding workflow for users, managing account activation cycles and checking constraints like email/phone uniqueness.
- **fed & fed_rpc**: Serves as the core Remote Procedure Call (RPC) gateway interface for encapsulating API requests, handling booking timelines, and delegating control flow to lower-level services.
- **mongo_db**: Operates the data persistence layer, establishing optimized database connections and handling read/write query executions for the platform.
- **fed_tool & util**: Provides shared cryptographic utilities, error-handling routines, configuration parsers, and geospatial location logic.

6.2 Non-UI Background Services

Beyond interactive components, the backend operates continuous background services ensuring operational integrity:

- **Rental Conflict Validator**: A real-time execution query intercepting and blocking overlapping rental timelines for the same vehicle ID.
- **Late Return Fee Processor**: A scheduled background worker that evaluates overdue return hour gaps and automatically appends penalty charges.
- **Automated Booking Reminder Driver**: A background cron job that triggers email/SMS alerts to customers before their contract starts or ends.
- **Live In-App Notification Pusher**: A persistent WebSocket worker thread instantly dispatching contextual alerts regarding booking confirmations and administrative alerts.
- **Low Availability Alert Sentinel**: A continuous inventory counter service that flashes critical warning flags on the Admin dashboard when specific vehicle model volumes drop below safe thresholds.

6.3 Technologies

- ReactJS
- React Native
- ASP.NET Core
- MongoDB
- GitHub
- Postman

6.4 Current Progress

- Requirements Analysis
- Literature Review

- Use Case Design
- Database Design (ERD)
- User Interface Design
- Authentication Module
- Google Authentication Integration
- Citizen ID Verification Design
- Booking Management Module
- Rental Conflict Validation Logic
- Rental Price Calculation Module
- Payment Integration Design
- Notification Service Design
- CRUD Functions for Users, Vehicles, and Bookings

7 Conclusion

This paper presented Motov, a hybrid motorbike rental platform integrating company-owned and partner-owned vehicles into a unified rental ecosystem. The proposed system supports customers, partners, and administrators through web and mobile applications. Future work includes recommendation systems, real-time vehicle tracking, and advanced analytics.

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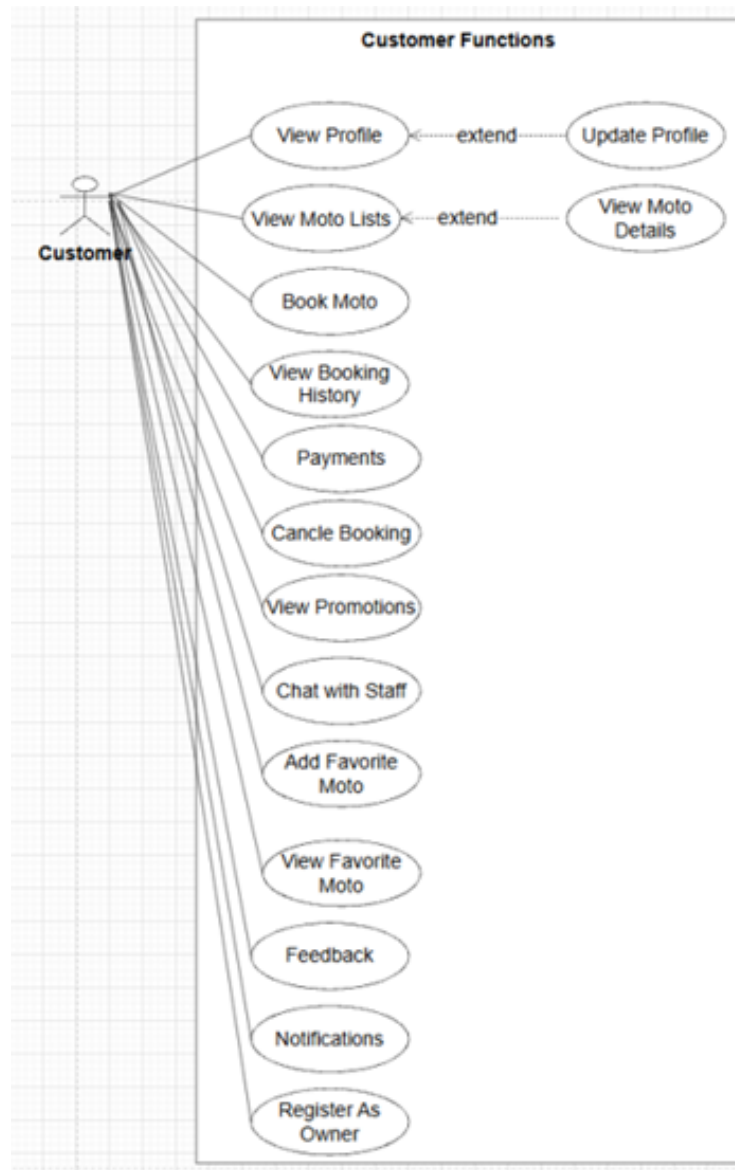
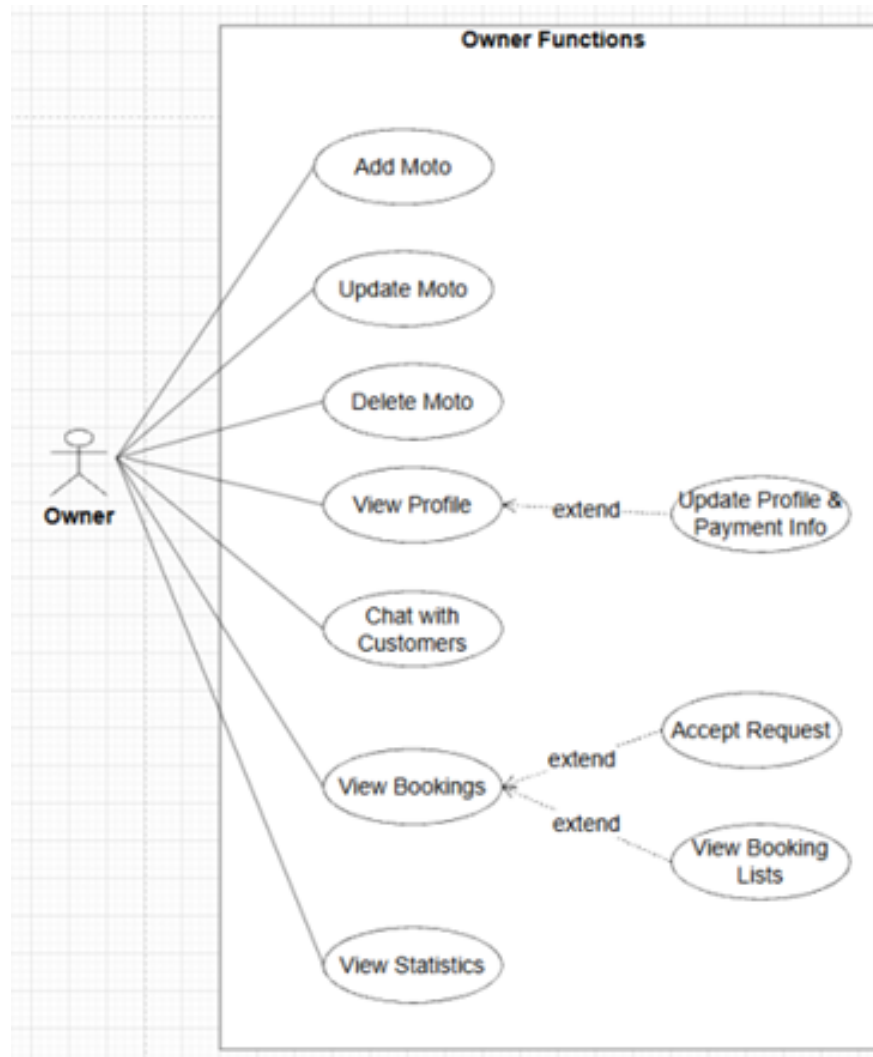


Fig. 2. Customer Use Case Diagram

**Fig. 3.** Owner Use Case Diagram

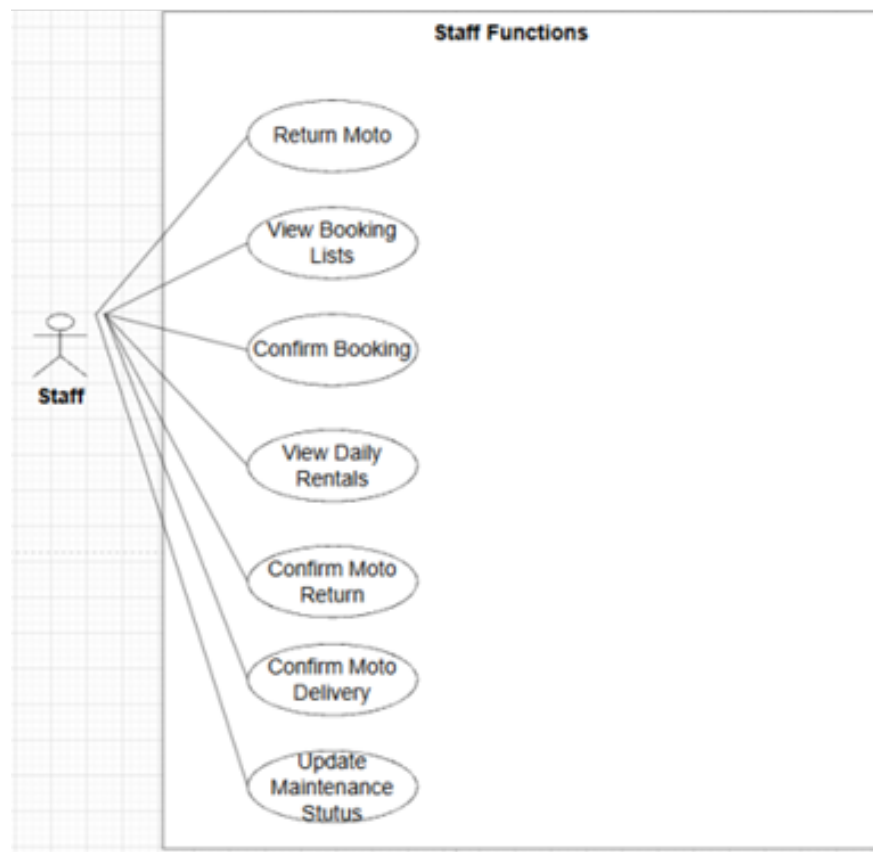


Fig. 4. Staff Use Case Diagram

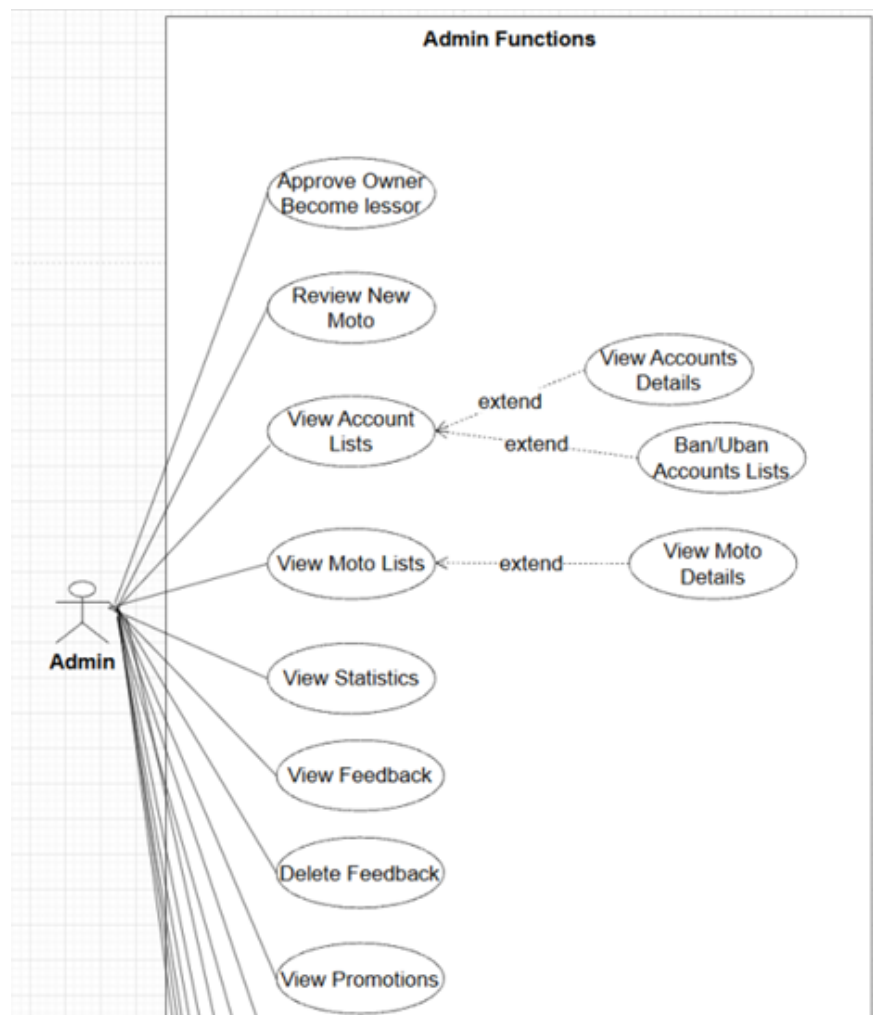


Fig. 5. Administrator Use Case Diagram (Part 1)

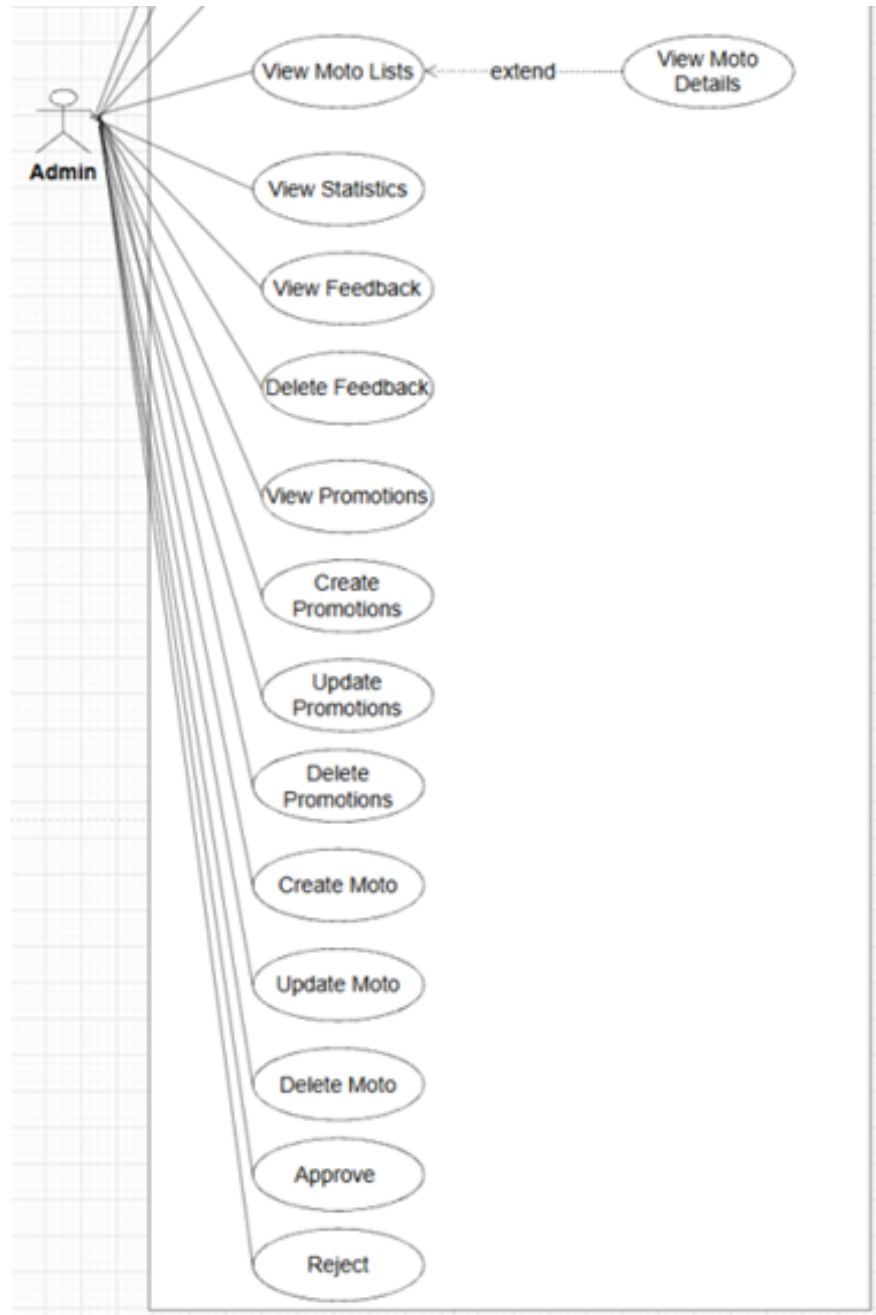


Fig. 6. Administrator Use Case Diagram (Part 2)

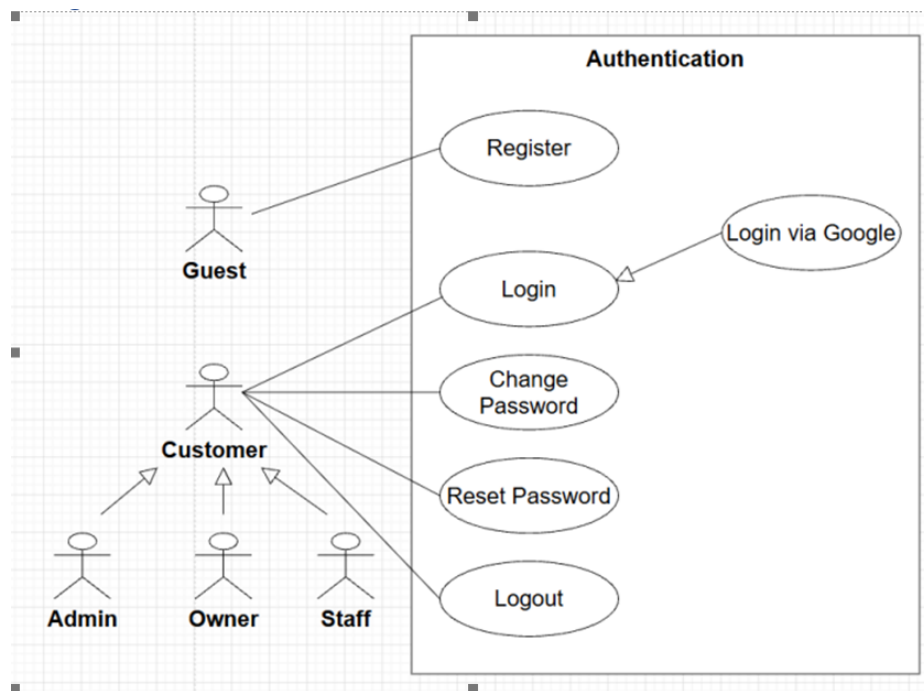


Fig. 7. Entity Relationship Diagram of the Motov system